



MATHEMATICS

Kenya Institute of Curriculum Development





ESSENCE STATEMENT

Mathematics is a vehicle of development and improvement of a country's economic development. By learning mathematics, learners develop an understanding of numbers, logical thinking skills and problem solving skills. Mathematics is applied in business, social and political worlds.

At this level mathematics will build on the competencies acquired by the learner in the early years of education. Learning mathematics will also enhance the learner's competencies in numeracy as a foundation of STEM at the higher levels of Education cycle. Mathematics is also a subject of enjoyment and excitement as it gives learners opportunities for creative work and fun.



General Learning Outcomes

By the end of Upper Primary the learner, should be able to:

- 1) Demonstrate mastery of number concepts by working out problems in day to day life.
- 2) Apply measurement skills to find solutions to problems in a variety of contexts.
- 3) Describe properties of geometrical shapes and spatial relationships in real life experiences.
- 4) *Collect, represent and analyze data to solve problems.***
- 5) *Analyze information using algebraic expressions in real life situations.***



Relating mathematics concepts to real life situations



Strand: Numbers:

Sub strands: Addition, Subtraction, Multiplication and Division

The curriculum design has indicated that the learners should work out questions involving real life situations. At this level, learners can be able to read and comprehend short sentences. The teacher should therefore involve the learners in word questions that depict day to day activities or real life situations. Abstract questions involving whole numbers, fractions, decimals or any other sub strands may not be of much help to the learners as they may not relate the concepts to everyday life.





The following are examples of word questions that the teacher may give the learners:

1. Amina harvested 276 bags of maize in one season. In the following season she harvested 405 bags of maize. How many bags of maize did she harvest in the two seasons altogether?
2. Joshua had 1890 metres of barbed wire. He used 738 metres to fence his shamba. How many metres of the wire was he left with?





3. Ouma bought 47 boxes of soap. Each box had 18 pieces of soap. How many pieces of soap were there altogether?

4. One thousand two hundred and forty eight tree seedlings were shared equally among 12 families. How many tree seedlings did each family get?





Strategies



In Early Years Education, the learners were taken through different strategies of working out questions under the various sub-strands. The strategies included:

- Counting on
- Counting forward
- Counting backward
- Number line
- Number family
- Breaking apart

These strategies may also be used at this level where learners have challenges.





Example

When 5 is subtracted from a number, the result is 7. What number was 5 subtracted from?

The numerical question is $\square - 5 = 7$

To work out $\square - 5 = 7$.

The missing number is made up of 5 and 7, therefore

$$5 + 7 = \square$$

Since $5 + 7 = 12$, then the missing number is 12

$$\square - 5 = 7$$





STRAND: Measurement

Measurement at this level includes **length, mass, volume, capacity, area, money and time**. When dealing with these sub strands, involve the learners in real life activities which they can relate with.

Example

1. A pencil is 15cm long. What is the total length of 8 similar pencils?
2. Four hundred and seventy litres of milk was packed in bottles each holding 10 litres. How many 10 litre bottles of milk were packed?





The Classroom Shop(money)

It is important for the teacher to compile a price list for the items in the shop. The teacher should guide the learners in the various shopping activities using real money or imitation money. The needs and wants in the sub strand on money may be discussed using the classroom shop depending on the items displayed. With guidance learners to discuss the money people pay to the County Government for provision of services as guided by the curriculum design. The money paid should not be referred to as tax at this level (grade 4).





DATA HANDLING: Data

Specific Learning outcomes

- ... By the end of the sub strand the learner should be able to:
 - a) represent data involving real life situations using frequency tables;



ALGEBRA: Use of letters

Specific Learning outcomes

**By the end of the sub strand the learner
should be able to:**

- a) represent the unknowns in real life
situation using letters;**



THANK YOU

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